



RIFT VALLEY UNIVERSITY



HAWASSA CAMPUS

DEPARTMENT OF ECONOMICS

Econometrics I Model Examination

Instruction: Choose the correct answer from the given alternatives!

1. All of the following methods are used in testing the individual significance of the parameter estimates of the OLS regression except

A. Standard error test B. F-test C. Confidence interval test D. Students t-test

Answer: **B**

2. The fitted regression equation is given by $Y = -12 + 0.5X$. What is the value of the residual at the point $X = 50$ and $Y = 70$?

A. 13 B. 42 C. 57 D. 29

Answer: **C**

3. One of the following is not a component of the BLUE property of the OLS estimators.

A. Linearity B. Unbiasedness C. Efficiency D. Maximum Variance

Answer: **D**

4. If the total sum of squares in a regression equation is 81, and the residual sum of squares is 25, what is the explained sum of squares?

A. 64 B. 56 C. 32 D. 18

Answer: **B**

5. One of the following is a problem in a regression which arises when some or all explanatory variables are perfectly linearly correlated with each other.

A. Heteroskedasticity B. Autocorrelation C. Multicollinearity D. Normality

Answer: **C**

6. Which of the following is a statistic that can be used to test hypothesis about a single population parameter?

A. F statistic B. t statistic C. χ^2 statistic D. Durbin Watson statistic

Answer: **B**

7. In a regression model with 10 observations and 2 regressors the degree of freedom is
- A. 2 B. 8 C. 7 D. 9

Answer: **C**

8. Which one of the following tests is appropriate to detect the presence of heteroskedasticity problem in a specified regression model?
- A. Darbin Watson d-statistic test C. Variance inflation factor
B. Breusch-Pagan-Godfrey Test D. Ramsey regression equation specification test

Answer: **B**

9. If the expected value of an estimator equals the true population parameter, the estimator is said to be
- A. Efficient B. Unbiased C. Biased D. Inefficient

Answer: **B**

10. If the residual sum of squares in a regression analysis is 35 and the explained sum of squares is 90, what is the coefficient of determination?
- A. 0.72 B. 0.55 C. 0.27 D. 3.75

Answer: **A**

11. Heteroskedasticity problem will arise in a regression model if
- A. different error terms have identical variances
B. different error terms do not have identical variances
C. the variance of dependent variable is not constant
D. the variance of the error term is constant regardless of sample size

Answer: **B**

12. The sum of the residuals is always
- A. ∞ B. 1 C. -1 D. 0

Answer: **D**

13. Given that $Y = f(X)$, a relationship between X and Y is said to be stochastic if
- A. Y assumes some probability for each value of X
B. for each value of the X there is only one corresponding value of Y
C. Y is not influenced by X at all
D. More than one corresponding values of Y exist for each value of X

Answer: **A**

14. Autocorrelation is said to be present in the specified regression model if the Darbin-Watson d-statistic is nearly close to

- A. 0 B. -2 C. 4 D. 2

Answer: **D**

15. In a regression analysis the expression BLUE stands for?

- A. Best Linear Unbiased Estimator. C. Bohr's Linear Unbiased Estimator.
B. Biased Linear Unit Estimator. D. Best Linear Unit Estimator.

Answer: **A**

16. The classical linear regression assumption that the correlation between any two error terms is zero denotes

- A. Serial Autocorrelation C. No multicollinearity
B. No autocorrelation D. Perfect multicollinearity

Answer: **B**

17. Which of the following is correct statement?

- A. The larger the error variance, the greater the variance of slope parameters.
B. The larger the error variance, the smaller the variance of slope parameters.
C. The smaller the error variance, the greater the variance of slope parameters.
D. The change in the error variance will have no significant influence the variance of slope parameters.

Answer: **A**

18. The variance inflation factor of 1 is an indication of.

- A. Perfect multicollinearity C. Moderate multicollinearity
B. High multicollinearity D. No multicollinearity

Answer: **D**

19. Given the linear regression function, $Y = \alpha + \beta X + U$, what is the assumption that the variance of the error or disturbance term is constant regardless of the value of X?

- A. Heteroskedasticity B. Autocorrelation C. Homoskedasticity D. No Autocorrelation

Answer: **C**

20. In a regression model with no explanatory variables the R^2 is equal to

- A. 1 B. 0 C. 0.5 D. -1

Answer: **B**

21. Suppose the researcher surveyed the same sample of households quarterly on repeated observations in the year 2022. The data collected by the researcher is categorized as a

- A. Cross sectional B. Time series C. Panel D. Pooled cross section

Answer: **C**

22. The normality of the error terms can be represented by

- A. Error terms with zero mean and constant variance
B. Error terms with zero mean and variable variance
C. Error mean of one and infinitely large variance
D. Error mean of one and constant variance

Answer: **A**

23. Model specification errors arise when

- A. Relevant variables are omitted from the specified model.
B. Irrelevant variables are neglected from the specified model
C. All relevant variables are included in the specified model
D. The specified model is non linear

Answer: **A**

24. In the linear regression model, $Y = \alpha + \beta X + U$, the variable 'X' is said to be a(an)

- A. Regressand/ Predictand C. Explanatory variable
B. Endogenous variable D. Dependent variable

Answer: **C**

25. Given the linear regression, $Y = \alpha + \beta X + U$, an R^2 of 0.90 indicates that

- A. 90% of the variations in Y is explained by the changes in X
B. 10% of the variations in Y is explained by the changes in X
C. 90% of the variations in Y is attributed to factors included in the disturbance term
D. 90% of the variations are left unexplained by the changes in X

Answer: **A**

26. What would be the consequences for the OLS estimator if heteroskedasticity is present in a regression model but ignored?

- A. The estimators will be biased C. The estimators will be non-linear
B. The estimators will have minimum variances D. The estimators will be inefficient

Answer: **D**

27. In testing the statistical significance of the OLS estimators, what is our decision if the calculated value of t is greater than its critical or table value?

- A. We accept the null and reject the alternative hypothesis
- B. We accept the alternative and reject the null hypothesis
- C. We reject both the null and alternative hypothesis
- D. We accept both the null and alternative hypothesis

Answer: **B**

28. The slope coefficient will be statistically significant at 5% level if

- A. P-value is less 0.05
- B. P-value is greater than 0.05
- C. P-value is 1
- D. P-value is indeterminate

Answer: **A**

29. Suppose $\hat{Y} = 128.5 + 2.88X$ is an estimated regression line from a sample of 20 observations and the standard error of the slope parameter is 0.85. If the critical value of t at 0.025 level of significance and 18 degree of freedom is 2.10, the 95% confidence interval for the slope of parameter is constructed as

- A. (1.09, 4.67)
- B. (2.01, 3.47)
- C. (-1.22, 2.54)
- D. (-12.21, 10.32)

Answer: **A**

30. As the sample size increases, the variance of the slope coefficient

- A. Increases
- B. Decreases
- C. stays constant
- D. is indeterminate

Answer: **B**

31. $E(U_i, U_j) = 0$, when $i \neq j$ is termed as,

- A. Heteroscedasticity
- B. No Auto-Correlation
- C. Multi-collinearity
- D. Homoscedasticity

Answer: **B**

32. In the two variable linear regression model the slope coefficient measures

- A. The percentage change in Y which the model predicts for a unit change in X
- B. The percentage change in X which the model predicts for a unit change in Y
- C. The percentage change in Y which is explained by changes in X
- D. The percentage change in Y which is unexplained by changes in X

Answer: **A**

33. Given the consumption function, **consumption** = $\alpha + \beta \text{income} + u_i$, how to interpret the slope parameter (β) if the its estimated value is 0.72?
- A. If disposable income increases by \$1, consumption expenditure increases by 72 cents.
 - B. If consumption expenditure increases by \$1, disposable income increases by 72 cents.
 - C. Other things being constant, a \$1 increase in a disposable income raises the consumption expenditure by 72 cents.
 - D. Other things being constant, a \$1 increase in the consumption expenditure raises the disposable income by 72 cents.

Answer: **C**

34. The data collected from individuals, businesses or groups at a given point in time denotes
- A. Time series data
 - C. Cross sectional data
 - B. Panel data
 - D. Pooled cross sectional data

Answer: **C**

35. Choose the correct one from the following.

A. $0 \leq R^2 \leq 1$ B. $0 < R^2 < 1$ C. $0 \geq R^2 \geq 1$ D. $0 > R^2 > 1$

Answer: **A**

36. Why do we include the error term in the econometrics model?

- A. To show the percentage of variation in the dependent variable that is explained by the included regressor(s)
- B. To determine the overall level significance of the model
- C. To justify that the specified model is non-stochastic
- D. To capture the effect of unobserved factors on the dependent variable

Answer: **D**

37. The formula of coefficient of determination is

A. $1 + \text{RSS}/\text{TSS}$ B. $1 - \text{RSS}/\text{ESS}$ C. $1 - \text{RSS}/\text{TSS}$ D. $1 + \text{ESS}/\text{TSS}$

Answer: **C**

38. “In the traditional learning experiment, the effect of practice on performance is investigated”. Performance in this context is a(an)

- A. Independent variable
- C. Dependent variable
- B. Extraneous variable
- D. Control variable

Answer: **C**

39. If we do not reject the null hypothesis, we conclude that:

- A. there is enough statistical evidence to infer that the alternative hypothesis is true
- B. there is not enough statistical evidence to infer that the alternative hypothesis is true
- C. there is enough statistical evidence to infer that the null hypothesis is true
- D. the test is statistically insignificant at whatever level of significance

Answer: **B**

40. The purpose of hypothesis testing is to:

- A. test how far the mean of a sample is from zero
- B. determine whether a statistical result is significant
- C. determine the appropriate value of the significance level
- D. derive the standard error of the data

Answer: **B**

41. Suppose $\hat{Y} = -9.96 + 0.74X_1 + 0.50X_2$ is an estimated regression line from a sample of 10 observations and the coefficient of variation (R^2) is 0.81. The calculated F value for this estimated model will be

- A. 12.2 B. 13.5 C. 15.6 D. 14.9

Answer: **D**

42. Consider the estimated equation: $\hat{Q} = 698.9 - 2.28P$
(10.4) (0.52)

The t-statistic for the slope is approximately

- A. 1.76 B. 67.20 C. 0.52 D. 4.38

Answer: **D**

43. The t-statistics is calculated by dividing of:

- A. The least squares estimator by its standard error
- B. The slope coefficient by one
- C. The estimator minus its hypothesized value of the standard error
- D. The slope coefficients by the coefficient of determination

Answer: **A**

44. White's test is used for the detection of

- A. Heteroscedasticity B. Multicollinearity C. Autocorrelation D. Normality

Answer: **A**

45. Which one is not the assumption of Guassi-Markovs OLS estimation?

- A. Zero covariance between error terms
- B. Equal variance of disturbance terms
- C. Perfect multicollinearity
- D. Zero mean value of disturbance terms

Answer: **C**

46. The most usually used method of estimation in econometrics is:

- A. General weighted least square method
- B. Maximum likelihood method
- C. Momentum method of estimation
- D. Ordinary least square method of estimation

Answer: **D**

47. In confidence of interval estimation, $\alpha = 5\%$, this means that the interval includes the true β with probability of:

- A. 5%
- B. 50%
- C. 0.5%
- D. 95%

Answer: **D**

48. The OLS estimator is derived by:

- A. Minimizing the sum of absolute residuals
- B. Dividing the sample size by the number of parameters in the model
- C. Minimizing the sum of squared residuals
- D. Minimizing the sum of residuals

Answer: **C**

49. Durbin Watson test is associated with:

- A. Multicollinearity
- B. Heteroscedasticity
- C. Autocorrelation
- D. Model specification error

Answer: **C**

50. To obtain the slope estimator using the least squares principle, you divide the

- A. Sample variance of X by the sample variance of Y.
- B. Sample covariance of X and Y by the sample variance of Y.
- C. Sample covariance of X and Y by the sample variance of X.
- D. Sample variance of X by the sample covariance of X and Y.

Answer: **C**

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